Chapter 9. Rigid Motions of Euclidean Space

Distance-preserving maps, orthogonal matrices and reflections

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The second half of the course begins here. Euclidean space is the vector space $\mathbb{R}^n$ with its distance; its "rigid" motions, which preserve lengths and distances, motivate the abstract inner product spaces, groups and forms of the following chapters.

9.1 The dot product on $\mathbb{R}^n$

Motivation. The dot product gives $\mathbb{R}^n$ its geometry: length, distance, angle and perpendicularity. Orthonormal bases and orthogonal complements are the basic tools, and the question $\dim W^\perp = n - \dim W$ points ahead to Section 10.3.

Definition 9.1.1 (Dot product)

For $X = (a_1, \ldots, a_n)^t$, $Y = (b_1, \ldots, b_n)^t \in \mathbb{R}^n$,

$$\langle X, Y \rangle = X^t \cdot Y = \sum_{i=1}^{n} a_i b_i$$

is the **dot product** on $\mathbb{R}^n$; $\mathbb{R}^n$ with the dot product is **Euclidean space**.

Exercise 9.1.2

(a) $(X, Y) \mapsto \langle X, Y \rangle$ is a bilinear form on $\mathbb{R}^n$ (Definition 6.1.1).
(b) $\langle X, Y \rangle = \langle Y, X \rangle$.
(c) $\langle X, X \rangle \geq 0$, with equality iff $X = 0$.

Solution.

(a) $X^t(aY + bZ) = aX^tY + bX^tZ$, and similarly in X.

(b) $\sum a_i b_i = \sum b_i a_i$.

(c) $\langle X, X \rangle = \sum a_i^2 \geq 0$, and a sum of squares of reals vanishes iff every $a_i = 0$. □

Exercise 9.1.3

If $\langle Y, X \rangle = 0$ for all $Y \in \mathbb{R}^n$, then $X = 0$.

Solution. Take $Y = X$ and use Exercise 9.1.2(c). □

The dot product gives $\mathbb{R}^n$ a geometric (topological) structure; studying it is **Euclidean geometry**.

Definition 9.1.4 (Norm)

$\|X\| = \sqrt{\langle X, X \rangle}$ is the (**Euclidean**) **norm**, or length, of X.

Exercise 9.1.5

For $X \in \mathbb{R}^n$, $c \in \mathbb{R}$:
(a) $\|X\| \geq 0$, and $\|X\| = 0$ iff $X = 0$.
(b) $\|cX\| = |c|\|X\|$.
(c) If $X \neq 0$, then $\left\| \frac{1}{\|X\|} X \right\| = 1$. A vector of norm 1 is a **unit vector**.

Solution.

(a) Exercise 9.1.2(c).

(b) $\|cX\|^2 = c^2\langle X, X\rangle$.

(c) By (b) with $c = 1/\|X\|$. □

Exercise 9.1.6 (Parallelogram law)

$\|X+Y\|^2 + \|X-Y\|^2 = 2(\|X\|^2 + \|Y\|^2)$.

Solution. Expand: $\|X \pm Y\|^2 = \|X\|^2 \pm 2\langle X, Y\rangle + \|Y\|^2$, and add. □

The angle $\theta \in [0, \pi]$ between X and Y is measured in the plane $\langle X\rangle \oplus \langle Y\rangle \cong \mathbb{R}^2$, where

$$\langle X, Y\rangle = \|X\|\|Y\| \cos\theta. \qquad (\angle)$$

Exercise 9.1.7

(a) (Cauchy–Schwarz) $|\langle X, Y\rangle| \le \|X\|\|Y\|$.
(b) (Triangle inequality) $\|X+Y\| \le \|X\| + \|Y\|$.

Solution.

(a) If $Y = 0$ both sides vanish. Otherwise $0 \le \|X - tY\|^2 = \|X\|^2 - 2t\langle X, Y\rangle + t^2\|Y\|^2$ for all $t \in \mathbb{R}$, so the discriminant is ≤ 0: $\langle X, Y\rangle^2 \le \|X\|^2\|Y\|^2$. Equality holds iff X, Y are linearly dependent.

(b) $\|X+Y\|^2 = \|X\|^2 + 2\langle X, Y\rangle + \|Y\|^2 \le (\|X\| + \|Y\|)^2$ by (a). □

So $\|X - Y\|$ is a distance (metric) on $\mathbb{R}^n$:

Exercise 9.1.8

(a) $\|X - X\| = 0$.
(b) $\|X - Y\| = \|Y - X\|$.
(c) $\|X - Y\| \le \|X - Z\| + \|Z - Y\|$.

Solution.

(a) Clear.

(b) Exercise 9.1.5(b) with $c = -1$.

(c) Exercise 9.1.7(b) applied to $(X - Z) + (Z - Y)$. □

Definition 9.1.9 (Perpendicularity)

(a) $X \perp Y$ (X, Y are **perpendicular**, **orthogonal**) if $\langle X, Y\rangle = 0$.
(b) For $S, T \subseteq \mathbb{R}^n$, $S \perp T$ if $\langle X, Y\rangle = 0$ for all $X \in S$, $Y \in T$.

Exercise 9.1.10

The following are equivalent: (1) $X \perp Y$; (2) $\|X+Y\|^2 = \|X\|^2 + \|Y\|^2$ (Pythagoras); (3) $\|X+Y\| = \|X-Y\|$.

Solution. $\|X \pm Y\|^2 = \|X\|^2 + \|Y\|^2 \pm 2\langle X, Y\rangle$. So (2) and (3) both say $\langle X, Y\rangle = 0$. □

Definition 9.1.11 (Orthogonal and orthonormal sets)

(a) Nonzero $X_1, \ldots, X_m \in \mathbb{R}^n$ with $X_i \perp X_j$ for $i \neq j$ form an **orthogonal subset**; if moreover all are unit vectors, an **orthonormal subset**.
(b) For $W \leq \mathbb{R}^n$, a basis $\mathfrak{B}$ of W is an **orthogonal basis** if it is an orthogonal subset, and an **orthonormal basis** if it is orthonormal. Normalizing (Exercise 9.1.5(c)) turns an orthogonal basis into an orthonormal one.

Exercise 9.1.12

(a) An orthogonal subset $\{X_1, \ldots, X_m\}$ of $\mathbb{R}^n$ is linearly independent.
(b) For $W \leq \mathbb{R}^n$, a basis $\{X_1, \ldots, X_m\}$ of W is orthonormal iff $\langle X_i, X_j \rangle = \delta_{ij}$.

Solution.

(a) If $\sum c_i X_i = 0$, then $0 = \langle \sum c_i X_i, X_j \rangle = c_j \|X_j\|^2$, so $c_j = 0$.

(b) This is the definition. □

Exercise 9.1.13

The following are orthonormal bases:
(a) the standard basis $\mathcal{E}$ of $\mathbb{R}^n$;
(b) $\{(1,0)^t, (0,-1)^t\}$ of $\mathbb{R}^2$;
(c) $\{(\cos\theta, \sin\theta)^t, (-\sin\theta, \cos\theta)^t\}$ of $\mathbb{R}^2$;
(d) $\{(\cos\theta, \sin\theta, 0)^t, (-\sin\theta, \cos\theta, 0)^t, (0,0,1)^t\}$ of $\mathbb{R}^3$;
(e) $\{\frac{1}{\sqrt{5}}(2,1)^t, \frac{1}{\sqrt{5}}(-1,2)^t\}$ of $\mathbb{R}^2$;
(f) $\{\frac{1}{\sqrt{2}}(-1,1,0)^t, \frac{1}{\sqrt{3}}(1,1,1)^t, \frac{1}{\sqrt{6}}(1,1,-2)^t\}$ of $\mathbb{R}^3$;
(g) $\{(\cos\theta, \sin\theta, 0, 0)^t, (-\sin\theta, \cos\theta, 0, 0)^t, (0,0,\cos\eta, \sin\eta)^t, (0,0,-\sin\eta, \cos\eta)^t\}$ of $\mathbb{R}^4$.

Solution. In each case the inner products of distinct vectors vanish and each vector has norm 1 (using $\cos^2 + \sin^2 = 1$). An orthonormal set of n vectors in $\mathbb{R}^n$ is independent (Exercise 9.1.12(a)), hence a basis. □

So orthonormal bases of $\mathbb{R}^n$ exist, and (f) shows how an orthogonal basis is normalized.

Definition 9.1.14 (Orthogonal complement)

For $S \subseteq \mathbb{R}^n$, $S^\perp = \{X \in \mathbb{R}^n : Y \perp X \text{ for all } Y \in S\}$ ("S perp"). For $W \leq \mathbb{R}^n$, $W^\perp$ is the **orthogonal complement** of W.

Observation 9.1.15

Let $S \subseteq \mathbb{R}^n$, $W \leq \mathbb{R}^n$, and $\mathfrak{C}$ a basis of W.
(a) $S^\perp$ is a subspace of $\mathbb{R}^n$.
(b) $S^\perp = \langle S \rangle^\perp$.
(c) $W^\perp = \mathfrak{C}^\perp$.

Proof.

(a) If $S \perp X$ and $S \perp Y$, then $S \perp X + Y$ and $S \perp cX$.

(b) $S \subseteq \langle S \rangle$ gives $\langle S \rangle^\perp \subseteq S^\perp$. Conversely, elements of $\langle S \rangle$ are linear combinations of vectors of S (Proposition 3.1.6), so a vector perpendicular to S is perpendicular to $\langle S \rangle$.

(c) The case $S = \mathfrak{C}$ of (b). ■

By (b) it suffices to consider subspaces, and by (c) $X \perp W$ iff X is perpendicular to a basis of W.

Exercise 9.1.16

Let $\mathfrak{B}$ be a basis of $\mathbb{R}^n$ and $W \leq \mathbb{R}^n$.
(a) $0^\perp = \mathbb{R}^n$ and $(\mathbb{R}^n)^\perp = \mathfrak{B}^\perp = 0$.
(b) $W \cap W^\perp = 0$.
(c) $W \leq (W^\perp)^\perp$.

Solution.

(a) Every X is perpendicular to 0; $(\mathbb{R}^n)^\perp = 0$ by Exercise 9.1.3, and $\mathfrak{B}^\perp = (\mathbb{R}^n)^\perp$ by Observation 9.1.15(c).

(b) $X \in W \cap W^\perp$ gives $\langle X, X\rangle = 0$.

(c) Each $X \in W$ is perpendicular to every vector of $W^\perp$. □

Question 9.1.17

If $W \leq \mathbb{R}^n$, is $W = (W^\perp)^\perp$?

There seems to be no direct proof of $(W^\perp)^\perp \leq W$; one is led to compute $\dim W^\perp$.

Example 9.1.18

For $0 \neq Y \in \mathbb{R}^n$, the linear map $\varphi_Y : \mathbb{R}^n \to \mathbb{R}$, $\varphi_Y(X) = \langle Y, X\rangle = Y^t X = L_{Y^t}(X)$, is surjective, since $\varphi_Y(Y) \neq 0$. By the dimension theorem,

$$\dim\langle Y\rangle^\perp = \dim\ker\varphi_Y = n - \dim\operatorname{im}\varphi_Y = n - 1.$$

So $\dim W^\perp = n - \dim W$ holds when $\dim W = 1$.

If $\dim W^\perp = n - \dim W$ held for all $W \leq \mathbb{R}^n$, then $\dim(W^\perp)^\perp = \dim W$, so $W = (W^\perp)^\perp$ by Exercise 9.1.16(c) (a *dimension argument*: $U \leq V$ and $\dim U = \dim V$ force $U = V$). And by Exercise 9.1.16(b) and Observation 7.6.4, $\mathbb{R}^n = W \oplus W^\perp$.

Question 9.1.19

If $W \leq \mathbb{R}^n$, do we always have $\dim W^\perp = n - \dim W$ and $\mathbb{R}^n = W \oplus W^\perp$?

The answer (yes) uses the homogeneous system $AX = 0$ and the rank theorem; it is given in §10.3.

9.2 Rigid motions of $\mathbb{R}^n$

Motivation. Rigid motions preserve distances. Using only the dimension theorem and the basic theorem on linear maps, we show that every rigid motion is a translation composed with a *linear* rigid motion; in particular it is a bijection.

Definition 9.2.1 (Rigid motion)

$M : \mathbb{R}^n \to \mathbb{R}^n$ is a **rigid motion** (**isometry**) of $\mathbb{R}^n$ if $\|M(X) - M(Y)\| = \|X - Y\|$ for all $X, Y \in \mathbb{R}^n$.

Example 9.2.2

For $Y \in \mathbb{R}^n$, the **translation** by Y, $T_Y(X) = X + Y$, is a rigid motion; T_Y is a bijection with $(T_Y)^{-1} = T_{-Y}$, and $T_0 = I$.

Example 9.2.3

For $R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, the **rotation** L_{R_θ} by θ about the origin is a linear rigid motion of $\mathbb{R}^2$ (Exercise 4.3.5). Abusing notation, write $L_{R_\theta} = R_\theta$; R_θ is a bijection, $(R_\theta)^{-1} = R_{-\theta}$, $R_0 = I$.

Example 9.2.4

$S = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ gives a linear rigid motion $L_S = S$ of $\mathbb{R}^2$ (Exercise 4.3.6), a bijection with $S^{-1} = S$. (It is a reflection; see §9.4.)

The columns of R_θ and S are the orthonormal bases of Exercise 9.1.13; this is explained in Observation 9.3.6.

Observation 9.2.5

A rigid motion M is injective.

Proof. $M(X) = M(Y)$ gives $\|X - Y\| = 0$, so $X = Y$. ■

Question 9.2.6

Is every rigid motion of $\mathbb{R}^n$ a bijection?

The answer (Corollary 9.2.12) needs the dimension theorem and the basic theorem of Chapter 5.

Exercise 9.2.7

(a) A composite of rigid motions is a rigid motion.
(b) The inverse of a bijective rigid motion is a rigid motion.

Solution.

(a) $\|MN(X) - MN(Y)\| = \|N(X) - N(Y)\| = \|X - Y\|$.

(b) Put $X = M(X')$, $Y = M(Y')$: $\|M^{-1}X - M^{-1}Y\| = \|X' - Y'\| = \|MX' - MY'\| = \|X - Y\|$. □

For a rigid motion M, $L = T_{-M(0)} \circ M$ is a rigid motion with $L(0) = 0$, and $M = T_{M(0)} \circ L$. We claim such an L is linear.

Observation 9.2.8

A rigid motion L of $\mathbb{R}^n$ with $L(0) = 0$ satisfies:
(a) $\|L(X)\| = \|X\|$;
(b) $\langle L(X), L(X)\rangle = \langle X, X\rangle$;
(c) $\langle L(X), L(Y)\rangle = \langle X, Y\rangle$;
(d) if $\mathfrak{B}$ is an orthonormal basis of $\mathbb{R}^n$, so is $L(\mathfrak{B})$.

Proof.

(a) Put $Y = 0$ in $\|L(X) - L(Y)\| = \|X - Y\|$.

(b) Square (a).

(c) Square $\|L(X)-L(Y)\| = \|X-Y\|$ and expand: $\langle LX, LX\rangle - 2\langle LX, LY\rangle + \langle LY, LY\rangle = \langle X, X\rangle - 2\langle X, Y\rangle + \langle Y, Y\rangle$; by (b), $\langle LX, LY\rangle = \langle X, Y\rangle$.

(d) If $\mathfrak{B} = \{X_i\}$, then $\langle L(X_i), L(X_j)\rangle = \langle X_i, X_j\rangle = \delta_{ij}$ by (c), so $L(\mathfrak{B})$ is an orthonormal set of n vectors, hence a basis (Exercise 9.1.12). ■

Observation 9.2.9

If the columns $[A]^1, \ldots, [A]^n$ of $A \in \mathfrak{M}_{n,n}(\mathbb{R})$ form an orthonormal basis of $\mathbb{R}^n$, then L_A is a rigid motion.

Proof. For $X = \sum a_i e_i$, $AX = \sum a_i Ae_i = \sum a_i [A]^i$, so $\|AX\|^2 = \sum_{i,j} a_i a_j \delta_{ij} = \sum a_i^2 = \|X\|^2$. Replacing X by $X - Y$ gives $\|AX - AY\| = \|X - Y\|$. ■

Observation 9.2.10

If a rigid motion L with $L(0) = 0$ fixes every vector of a basis $\mathfrak{B} = \{X_i\}$ of $\mathbb{R}^n$, then $L = I$.

Proof. For $X \in \mathbb{R}^n$, $\langle X_i, L(X)\rangle = \langle L(X_i), L(X)\rangle = \langle X_i, X\rangle$ by Observation 9.2.8(c). So $L(X) - X \in \mathfrak{B}^\perp = 0$ (Exercise 9.1.16(a)). ■

Theorem 9.2.11

(a) A rigid motion L of $\mathbb{R}^n$ with $L(0) = 0$ is linear.
(b) Hence every rigid motion M of $\mathbb{R}^n$ is a translation composed with a linear rigid motion.

Proof.

(a) We look for A with $L = L_A$. Put $X_j = L(e_j)$ and $A = (X_1, \ldots, X_n)$, so $Ae_j = X_j$. By Observation 9.2.8(d), $\{X_j\}$ is an orthonormal basis, so A is invertible (basic theorem, Chapter 5) and L_A is invertible. Consider $M' = (L_A)^{-1} \circ L$. Then $M'(0) = 0$ and $M'(e_j) = A^{-1}X_j = e_j$. M' is a rigid motion by Observation 9.2.9 and Exercise 9.2.7, so $M' = I$ by Observation 9.2.10, i.e. $L = L_A$.

(b) With $L = T_{-M(0)} \circ M$, L is linear by (a) and $M = T_{M(0)} \circ L$. ■

Second proof of (a). By Observation 9.2.8(c), for all j,

$$\langle L(e_j), L(X+Y)\rangle = \langle e_j, X+Y\rangle = \langle e_j, X\rangle + \langle e_j, Y\rangle = \langle L(e_j), L(X) + L(Y)\rangle.$$

Since $L(\mathcal{E})$ is a basis (Observation 9.2.8(d)), $L(X+Y) - L(X) - L(Y) \in L(\mathcal{E})^\perp = 0$. Likewise $\langle L(e_j), L(cX)\rangle = c\langle e_j, X\rangle = \langle L(e_j), cL(X)\rangle$ gives $L(cX) = cL(X)$. ■

The second proof is shorter but offers no motivation; the first shows why the basic theorem and Exercise 9.2.7 matter.

Corollary 9.2.12

Every rigid motion of $\mathbb{R}^n$ is a bijection.

Proof. A linear rigid motion is injective (Observation 9.2.5), hence bijective by the dimension theorem. Translations are bijections, so by Theorem 9.2.11(b) every rigid motion is a bijection. ■

Exercise 9.2.13

Let $\mathcal{A}(\mathbb{R}^n)$ be the set of rigid motions of $\mathbb{R}^n$. (a) If $L, M \in \mathcal{A}(\mathbb{R}^n)$, then $L \circ M \in \mathcal{A}(\mathbb{R}^n)$.
(b) If $L \in \mathcal{A}(\mathbb{R}^n)$, then $L^{-1} \in \mathcal{A}(\mathbb{R}^n)$.

Solution. Exercise 9.2.7, with Corollary 9.2.12 for (b). □

A set with these two properties (and containing I) is a **group**; groups are studied from Chapter 11.

Exercise 9.2.14

Let $L_A, L_B \in \mathcal{A}(\mathbb{R}^n)$ and $Y, Z \in \mathbb{R}^n$.
(a) If $T_Y \circ L_A = T_Z \circ L_B$, then $A = B$ and $Y = Z$. (So every element of $\mathcal{A}(\mathbb{R}^n)$ is uniquely $T_Y \circ L_A$.)
(b) $L_A \circ T_Y = T_{AY} \circ L_A$.

Solution.

(a) Evaluate at 0: $Y = Z$. Then $L_A = L_B$, so $A = B$.

(b) $A(X + Y) = AX + AY$. □

Since translations are well understood, from now on we study *linear* rigid motions.

9.3 Orthogonal operators and orthogonal matrices

Motivation. Linear rigid motions are the orthogonal operators. They are characterised by preserving lengths, dot products or orthonormal bases, and their matrices by $A^tA = I$. Those with determinant 1 form the special orthogonal group.

Definition 9.3.1 (Orthogonal operators and matrices)

(a) A *linear* rigid motion of $\mathbb{R}^n$ is a (real) **orthogonal operator**; if $L = L_A$ is one, the real $n \times n$ matrix A is an **orthogonal matrix**.
(b) The **orthogonal group** on $\mathbb{R}^n$ is $\mathbf{O}(\mathbb{R}^n) = \{L \in \mathfrak{L}(\mathbb{R}^n, \mathbb{R}^n) : \|L(X) - L(Y)\| = \|X - Y\| \ \forall X, Y\}$, and

$$\mathbf{O}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{R}) : \|AX - AY\| = \|X - Y\| \ \forall X, Y \in \mathbb{R}^n\}.$$

Corollary 9.3.2

For every rigid motion M of $\mathbb{R}^n$ there are unique $A \in \mathbf{O}(n)$, $B \in \mathbb{R}^n$ with $M(X) = AX + B$.

Proof. In Theorem 9.2.11, $M = T_{M(0)} \circ L$ with $L = L_A$ linear; put $B = M(0)$. Uniqueness is Exercise 9.2.14(a). ■

Exercise 9.3.3

(a) If $L, M \in \mathbf{O}(\mathbb{R}^n)$, then $L \circ M \in \mathbf{O}(\mathbb{R}^n)$; if $A, B \in \mathbf{O}(n)$, then $AB \in \mathbf{O}(n)$.
(b) If $L \in \mathbf{O}(\mathbb{R}^n)$, then $L^{-1} \in \mathbf{O}(\mathbb{R}^n)$; if $A \in \mathbf{O}(n)$, then $A^{-1} \in \mathbf{O}(n)$.

Solution. Composites and inverses of linear rigid motions are linear rigid motions (Exercise 9.2.13); $L_A L_B = L_{AB}$ and $(L_A)^{-1} = L_{A^{-1}}$. □

Observation 9.3.4

For linear $L : \mathbb{R}^n \to \mathbb{R}^n$ the following are equivalent:
(1) $\|LX - LY\| = \|X - Y\|$ for all X, Y (i.e. $L \in \mathbf{O}(\mathbb{R}^n)$);
(2) $\|LX\| = \|X\|$ for all X;
(3) $\langle LX, LX\rangle = \langle X, X\rangle$ for all X;
(4) $\langle LX, LY\rangle = \langle X, Y\rangle$ for all X, Y;
(5) if $\mathfrak{B}$ is an orthonormal basis of $\mathbb{R}^n$, so is $L(\mathfrak{B})$.

Proof. (1)⇒(2): $Y = 0$. (2)⇒(1): replace X by $X - Y$, using $L(X - Y) = LX - LY$. (2)⇔(3): square. (3)⇒(4): replace X by $X - Y$ (or $X + Y$) and expand, as in Observation 9.2.8(c). (4)⇒(3): clear. (4)⇒(5): $\langle LX_i, LX_j\rangle = \langle X_i, X_j\rangle = \delta_{ij}$. (5)⇒(2): take an orthonormal basis $\mathfrak{B} = \{X_i\}$ (e.g. $\mathcal{E}$) and $X = \sum a_i X_i$. Then $\|X\|^2 = \sum a_i^2$, and since $L\mathfrak{B}$ is orthonormal, $\|LX\|^2 = \|\sum a_i LX_i\|^2 = \sum a_i^2$. ■

So orthogonal operators preserve distance, length, dot product and orthonormal bases.

Exercise 9.3.5

The conditions of Observation 9.3.4 are also equivalent to
(2)′ if $\|X\| = 1$, then $\|LX\| = 1$ (L preserves unit vectors);
(5)′ some orthonormal basis $\mathfrak{B}$ has $L\mathfrak{B}$ orthonormal.

Solution. (2)⇒(2)′ and (5)⇒(5)′ are clear. (2)′⇒(2): for $X \neq 0$, $\|LX\| = \|X\| \left\|L(\frac{X}{\|X\|})\right\| = \|X\|$. (5)′⇒(2): the last step of the proof of Observation 9.3.4 uses only one orthonormal basis. □

Observation 9.3.6

For $A \in \mathfrak{M}_{n,n}(\mathbb{R})$ the following are equivalent:
(1) $\|AX - AY\| = \|X - Y\|$ for all X, Y (i.e. $A \in \mathbf{O}(n)$);
(2) $\|AX\| = \|X\|$;
(3) $\langle AX, AX\rangle = \langle X, X\rangle$;
(4) $\langle AX, AY\rangle = \langle X, Y\rangle$;
(5) the columns $\{[A]^1, \ldots, [A]^n\}$ form an orthonormal basis of $\mathbb{R}^n$;
(6) $A^tA = I = AA^t$, i.e. $A^{-1} = A^t$;
(7) $A^t \in \mathbf{O}(n)$;
(8) the rows $\{[A]_1, \ldots, [A]_n\}$ form an orthonormal basis of $\mathfrak{M}_{1,n}(\mathbb{R})$.

Proof. (1)⇔(2)⇔(3)⇔(4)⇔(5): Observation 9.3.4 for $L = L_A$ with $\mathfrak{B} = \mathcal{E}$, using $[A]^i = Ae_i$.

(5)⇔(6): the (i, j) entry of A^tA is the dot product of column i and column j of A; so $A^tA = I$ iff $\langle [A]^i, [A]^j\rangle = \delta_{ij}$. For square matrices $A^tA = I$ implies $AA^t = I$.

(6)⇒(7) [Supplement]: $(A^t)^tA^t = AA^t = I$, so A^t satisfies (6). (7)⇒(6): apply (6) to A^t.

(7)⇔(8) [Supplement]: the rows of A are the transposed columns of A^t; apply (5) to A^t. ■

Second proof of (4)⇔(6). (4) says $X^tY = (AX)^tAY = X^t(A^tA)Y = X^tIY$ for all X, Y, which by Exercise 9.3.7 means $A^tA = I$. The converse reads the computation backwards. ■

Exercise 9.3.7

If $B, C \in \mathfrak{M}_{m,n}(F)$ and $X^tBY = X^tCY$ for all $X \in F^m$, $Y \in F^n$, then $B = C$.

Solution. $e_i^tBe_j = B_{ij}$; take $X = e_i$, $Y = e_j$. □

So finding $n \times n$ orthogonal matrices is the same as finding orthonormal bases of $\mathbb{R}^n$, and

$$\mathbf{O}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{R}) : A^t A = I\}.$$

With $(L_A)^t = L_{A^t}$, $\mathbf{O}(\mathbb{R}^n) = \{L \in \mathfrak{L}(\mathbb{R}^n, \mathbb{R}^n) : L^t \circ L = I\}$; defining the transpose of a general linear map is one of the goals of later chapters.

Exercise 9.3.8

If $A \in \mathbf{O}(n)$, then $\det A = \pm 1$.

Solution. $1 = \det(A^t A) = \det(A^t) \det A = (\det A)^2$. □

Definition 9.3.9 (Special orthogonal group)

$\mathbf{SO}(n) = \{A \in \mathbf{O}(n) : \det A = 1\}$ and $\mathbf{SO}(\mathbb{R}^n) = \{L \in \mathbf{O}(\mathbb{R}^n) : \det L = 1\}$.

In geometry, L_A with $A \in \mathbf{SO}(n)$ is **orientation preserving**, and with $A \in \mathbf{O}(n) - \mathbf{SO}(n)$ **orientation reversing**.

Exercise 9.3.10

(a) $I \in \mathbf{SO}(n)$.
(b) If $A, B \in \mathbf{SO}(n)$, then $AB \in \mathbf{SO}(n)$.
(c) If $A \in \mathbf{SO}(n)$, then $A^{-1} \in \mathbf{SO}(n)$.

Solution. Exercise 9.3.3 together with $\det(AB) = \det A \det B$ and $\det A^{-1} = (\det A)^{-1}$. □

Exercise 9.3.11

Fix $S \in \mathbf{O}(n) - \mathbf{SO}(n)$.
(a) Indeed $\mathbf{O}(n) - \mathbf{SO}(n) \neq \emptyset$.
(b) $\lambda_S : \mathbf{SO}(n) \to S \cdot \mathbf{SO}(n) = \{SA : A \in \mathbf{SO}(n)\}$, $\lambda_S(A) = SA$, is a bijection.
(c) $\mathbf{O}(n) - \mathbf{SO}(n) = S \cdot \mathbf{SO}(n)$.
(d) $\mathbf{O}(n) = \mathbf{SO}(n) \amalg (S \cdot \mathbf{SO}(n))$ (disjoint union).

Solution.

(a) $\mathrm{diag}(-1, 1, \ldots, 1)$.

(b) $SA = SB$ implies $A = B$; surjective by definition.

(c) If $R \in \mathbf{O}(n) - \mathbf{SO}(n)$, then $S^{-1}R \in \mathbf{SO}(n)$ ($\det = (-1)(-1)$), so $R = S(S^{-1}R)$; conversely $\det(SA) = -1$.

(d) Combine (c) with $\mathbf{SO}(n) \cap (\mathbf{O}(n) - \mathbf{SO}(n)) = \emptyset$. □

This is the first example of a coset decomposition (Chapter 12).

9.4 Reflections

Motivation. Reflecting across the hyperplane $\langle Y \rangle^\perp$ is the simplest orthogonal operator besides the identity. In an orthonormal basis starting with $Y/\|Y\|$ its matrix is $\mathrm{diag}(-1, 1, \ldots, 1)$. This "change to a good orthonormal frame" is the way of thinking used for the rest of the course.

Example 9.4.1

For $X, Y \in \mathbb{R}^3$, $Y \neq 0$, write $X = aY + (X - aY)$ with $(X - aY) \perp Y$; then $a\langle Y, Y\rangle = \langle X, Y\rangle$, so $a = \frac{\langle X,Y\rangle}{\langle Y,Y\rangle}$. Reflecting X in the plane $\langle Y\rangle^\perp$ gives $X - \frac{2\langle X,Y\rangle}{\langle Y,Y\rangle}Y$.

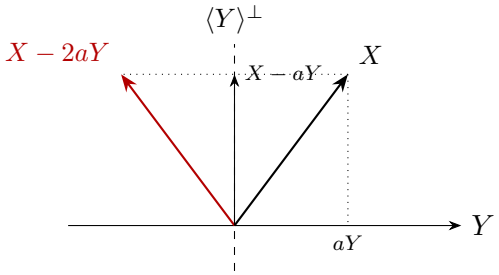


Figure. *Reflection of X in the hyperplane $\langle Y\rangle^\perp$.*

Definition 9.4.2 (Reflection)

For $0 \neq Y \in \mathbb{R}^n$, $S_Y : \mathbb{R}^n \to \mathbb{R}^n$,

$$S_Y(X) = X - \frac{2\langle X, Y\rangle}{\langle Y, Y\rangle}Y,$$

is the **reflection** (**symmetry**) with respect to Y.

Remark 9.4.3

The "reflection in the x-axis" of $\mathbb{R}^2$ is $S_{(0,1)^t}$ in this terminology.

Exercise 9.4.4

For $0 \neq Y \in \mathbb{R}^n$:
(a) $S_Y = S_{cY}$ for $0 \neq c \in \mathbb{R}$;
(b) $S_Y(Y) = -Y$;
(c) if $X \perp Y$, i.e. $X \in \langle Y\rangle^\perp$, then $S_Y(X) = X$.

Solution.
(a) $\frac{\langle X,cY\rangle}{\langle cY,cY\rangle}cY = \frac{c^2\langle X,Y\rangle}{c^2\langle Y,Y\rangle}Y$.
(b) $Y - 2Y = -Y$.
(c) $\langle X, Y\rangle = 0$. □

S_Y is visibly linear, and preserves lengths, so $S_Y \in \mathbf{O}(\mathbb{R}^n)$; one may also check this by computation:

Exercise 9.4.5

$S_Y \in \mathbf{O}(\mathbb{R}^n)$.

Solution. S_Y is linear because $X \mapsto \langle X, Y\rangle$ is. And $\|S_Y X\|^2 = \|X\|^2 - \frac{4\langle X,Y\rangle^2}{\langle Y,Y\rangle} + \frac{4\langle X,Y\rangle^2}{\langle Y,Y\rangle^2}\langle Y, Y\rangle = \|X\|^2$. Apply Observation 9.3.4. □

The way of thinking. By Example 9.1.18, $\langle Y\rangle \cap \langle Y\rangle^\perp = 0$ and $\dim\langle Y\rangle^\perp = n - 1$, so $\mathbb{R}^n = \langle Y\rangle \oplus \langle Y\rangle^\perp$ (Observation 7.6.4). If $\mathfrak{C} = \{X_2, \ldots, X_n\}$ is an ordered basis of $\langle Y\rangle^\perp$, then $\mathfrak{B} =$

$\{Y, X_2, \ldots, X_n\}$ is an ordered basis of $\mathbb{R}^n$. Exercise 9.4.4(b),(c) say $S_Y(Y) = -Y$ and $S_Y|_{\langle Y\rangle^\perp} = I$, which determine S_Y uniquely (linear extension theorem), and

$$[S_Y]_{\mathfrak{B}}^{\mathfrak{B}} = \operatorname{diag}(-1, 1, \ldots, 1).$$

Exercise 9.4.6

For $0 \neq Y \in \mathbb{R}^n$, $(S_Y)^2 = I$ and $(S_Y)^{-1} = S_Y$.

Solution. $[S_Y^2]_{\mathfrak{B}}^{\mathfrak{B}} = \operatorname{diag}(-1, 1, \ldots, 1)^2 = I$. □

Exercise 9.4.7

For $0 \neq Y \in \mathbb{R}^n$, $\det(S_Y) = -1$.

Solution. $\det S_Y = \det[S_Y]_{\mathfrak{B}}^{\mathfrak{B}} = \det \operatorname{diag}(-1, 1, \ldots, 1) = -1$. □

As a matrix, $S_Y = L_A$ for a unique $A \in \mathbf{O}(n)$, and

$$A = [S_Y]_{\mathcal{E}}^{\mathcal{E}} = [I]_{\mathcal{E}}^{\mathfrak{B}} \cdot [S_Y]_{\mathfrak{B}}^{\mathfrak{B}} \cdot [I]_{\mathfrak{B}}^{\mathcal{E}} = [I]_{\mathcal{E}}^{\mathfrak{B}} \cdot \operatorname{diag}(-1, 1, \ldots, 1) \cdot [I]_{\mathfrak{B}}^{\mathcal{E}}.$$

$[I]_{\mathfrak{B}}^{\mathcal{E}}$ (columns the vectors of $\mathfrak{B}$) is in $\mathbf{O}(n)$ iff $\mathfrak{B}$ is an orthonormal basis (Observation 9.3.6(5)). For that we may assume $\|Y\| = 1$ (Exercise 9.4.4(a)); then $\mathfrak{B}$ is orthonormal iff $\mathfrak{C}$ is an orthonormal basis of $\langle Y\rangle^\perp$.

Question 9.4.8

Does every $W \leq \mathbb{R}^n$ have an orthonormal basis?

Yes (Gram–Schmidt, next chapter). With $\mathfrak{C}$ orthonormal, all three factors above lie in $\mathbf{O}(n)$: the identity happens inside the orthogonal group. (Matrices related by $B = U^{-1}AU$ with $U \in \mathbf{O}(n)$ are *orthogonally similar*.) Thinking of orthonormal bases as orthonormal coordinate frames: in the frame $\mathfrak{B}$, the reflection S_Y flips one axis (the Y-axis) and fixes all the others. This is the way of thinking we will use repeatedly.

9.5 $\mathbf{O}(2)$ and $\mathbf{SO}(2)$

Motivation. Every orthogonal operator of the plane is a rotation R_θ or a reflection S_θ, according to its determinant. The composition rules for these, in particular "flip, rotate, flip back = rotate backwards", complete the description of $\mathbf{O}(2)$.

Example 9.5.1

$\mathbf{O}(1) = \{1, -1\}$ and $\mathbf{SO}(1) = \{1\}$.

Theorem 9.5.2

The elements of $\mathbf{SO}(2)$ are exactly the 2-dimensional rotations:

$$\mathbf{SO}(2) = \left\{ \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} : 0 \leq \theta < 2\pi \right\} = \left\{ \begin{pmatrix} x & -y \\ y & x \end{pmatrix} : x, y \in \mathbb{R},\ x^2 + y^2 = 1 \right\}.$$

High-school proof. Let $A = \left(\begin{smallmatrix} x & z \\ y & w \end{smallmatrix}\right) \in \mathbf{SO}(2)$: $x^2 + y^2 = 1$, $z^2 + w^2 = 1$, $xz + yw = 0$, $xw - yz = 1$. Write $(x, y) = (\cos\theta, \sin\theta)$ and $(z, w) = (\sin\eta, \cos\eta)$. Then $\sin(\theta + \eta) = xz + yw = 0$ and $\cos(\theta + \eta) = xw - yz = 1$, so $\theta + \eta \in 2\pi\mathbb{Z}$. Hence $z = \sin\eta = -\sin\theta$ and $w = \cos\eta = \cos\theta$. ■

Freshman's proof. If $y = 0$, then $x = \pm 1$; $xz = 0$ gives $z = 0$, and $xw = 1$ gives $w = x$: $A = \pm I$, of the stated form. If $y \neq 0$, $w = -\frac{xz}{y}$, and $1 = z^2 + w^2 = \frac{z^2(y^2+x^2)}{y^2} = \frac{z^2}{y^2}$, so $z = \pm y$. If $z = y$, then $w = -x$ and $xw - yz = -(x^2 + y^2) = -1$, impossible; so $z = -y$ and $w = x$. ■

Sophomore's proof. An orthogonal matrix is an orthonormal basis (Observation 9.3.6(5)). Its first vector is $(\cos\theta, \sin\theta)^t$, and the second, a unit vector perpendicular to it, is $(-\sin\theta, \cos\theta)^t$ or $(\sin\theta, -\cos\theta)^t$. The determinants are 1 and -1, so only the first lies in $\mathbf{SO}(2)$. ■

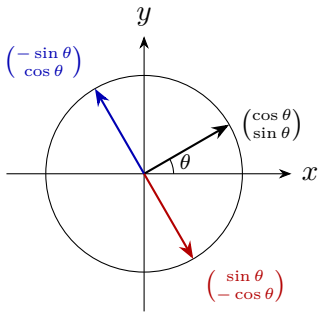


Figure. *The second vector of an orthonormal basis of $\mathbb{R}^2$ is one of two choices: determinant $+1$ (rotation) or -1 (reflection).*

The sophomore's proof describes all of $\mathbf{O}(2)$:

Corollary 9.5.3

$$\mathbf{O}(2) = \mathbf{SO}(2) \amalg \left\{ \begin{pmatrix} x & y \\ y & -x \end{pmatrix} : x, y \in \mathbb{R},\ x^2 + y^2 = 1 \right\}.$$

Proof. By Exercise 9.3.11, $\mathbf{O}(2) - \mathbf{SO}(2) = S \cdot \mathbf{SO}(2)$ for any $S \in \mathbf{O}(2) - \mathbf{SO}(2)$. With $S = \left(\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, the elements are $S\left(\begin{smallmatrix} x & -y \\ y & x \end{smallmatrix}\right) = \left(\begin{smallmatrix} -x & y \\ y & x \end{smallmatrix}\right)$; replace x by $-x$. ■

Write $R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ and $S_\theta = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$.

Question 9.5.4

Is S_θ a reflection of $\mathbb{R}^2$?

Question 9.5.5

For reflections S_Y, S_Z of $\mathbb{R}^2$, $\det(S_Y S_Z) = (-1)(-1) = 1$, so $S_Y \circ S_Z = R_\theta$ for some θ (Theorem 9.5.2). How do we find θ?

Observation 9.5.6

For $\theta \in \mathbb{R}$ and $Y_\theta = (-\sin\frac{\theta}{2}, \cos\frac{\theta}{2})^t$, $S_\theta = S_{Y_\theta}$. In particular S_θ is a reflection.

Proof. Both are linear, so compare them on e_1, e_2. With $s = \sin\frac{\theta}{2}$, $c = \cos\frac{\theta}{2}$: $S_{Y_\theta}(e_1) = e_1 - 2\langle e_1, Y_\theta\rangle Y_\theta = e_1 + 2s(-s, c)^t = (1-2s^2, 2sc)^t = (\cos\theta, \sin\theta)^t = S_\theta e_1$, and $S_{Y_\theta}(e_2) = e_2 - 2c(-s, c)^t = (2sc, 1-2c^2)^t = (\sin\theta, -\cos\theta)^t = S_\theta e_2$ (half-angle formulas). ■

Theorem 9.5.7

An orthogonal operator L of $\mathbb{R}^2$ is a rotation or a reflection: a rotation if $\det L = 1$, a reflection if $\det L = -1$.

Proof. $\mathbf{O}(2) = \mathbf{SO}(2) \amalg (\mathbf{O}(2) - \mathbf{SO}(2)) = \{R_\theta\} \amalg \{S_\theta\}$ (Corollary 9.5.3), and each S_θ is a reflection (Observation 9.5.6). ■

Example 9.5.8

(a) For a rotation R and a reflection S of $\mathbb{R}^2$, $R \circ S$ and $S \circ R$ are reflections (determinant -1).
(b) For reflections S, S', $S \circ S'$ is a rotation (determinant 1).

By the addition formulas $R_\theta \circ R_\eta = R_{\theta+\eta}$ and $(R_\theta)^{-1} = R_{-\theta}$; by Exercise 9.4.6, $(S_\theta)^2 = I$ and $(S_\theta)^{-1} = S_\theta$. The last rule:

Observation 9.5.9

For a reflection S of $\mathbb{R}^2$ and $\theta \in \mathbb{R}$: $S^{-1} \circ R_\theta \circ S = (R_\theta)^{-1} = R_{-\theta}$, i.e. $R_\theta \circ S = S \circ R_{-\theta}$, or $(S \circ R_\theta)^2 = I$.

Proof. $S \circ R_\theta$ is a reflection (Example 9.5.8), so $(S \circ R_\theta)^2 = I$ by Exercise 9.4.6. Thus $SR_\theta SR_\theta = I$, i.e. $SR_\theta S = (R_\theta)^{-1} = R_{-\theta}$, and $S = S^{-1}$. ■

"Flip, rotate, flip back = rotate backwards."

Example 9.5.10

Fix a reflection S of $\mathbb{R}^2$. By Exercise 9.3.11, $\mathbf{O}(2) - \mathbf{SO}(2) = S \cdot \mathbf{SO}(2)$, so every reflection is uniquely $S \circ R_\theta$. Products of rotations and reflections reduce to $R_\theta R_\eta$, $(SR_\theta)R_\eta$, $R_\theta(SR_\eta)$ and $(SR_\theta)(SR_\eta)$, and Observation 9.5.9 lets us move S past rotations; e.g. $(S \circ R_\theta) \circ (S \circ R_\eta) = S \circ (S \circ R_{-\theta}) \circ R_\eta = R_{\eta-\theta}$.

Example 9.5.11

Answer to Question 9.5.5: find θ, η with $S_Y = S_\theta$, $S_Z = S_\eta$, i.e. $\frac{1}{\|Y\|}Y = \pm(-\sin\frac{\theta}{2}, \cos\frac{\theta}{2})^t$ and $\frac{1}{\|Z\|}Z = \pm(-\sin\frac{\eta}{2}, \cos\frac{\eta}{2})^t$ (Observation 9.5.6, Exercise 9.4.4(a)). With $S = \left(\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right) = S_\pi$ we have $S \circ R_{\pi-\theta} = S_\theta$, so by Observation 9.5.9

$$S_\theta \circ S_\eta = (S \circ R_{\pi-\theta}) \circ (S \circ R_{\pi-\eta}) = S^2 \circ R_{\theta-\pi} \circ R_{\pi-\eta} = R_{\theta-\eta}.$$

Exercise 9.5.12

For $Y = (-1, 1)^t$, $Z = (-1, \sqrt{3})^t$, find θ with $S_Y \circ S_Z = R_\theta$.

Solution. $\frac{1}{\|Y\|}Y = (-\sin\frac{\pi}{4}, \cos\frac{\pi}{4})^t$ and $\frac{1}{\|Z\|}Z = (-\frac{1}{2}, \frac{\sqrt{3}}{2})^t = (-\sin\frac{\pi}{6}, \cos\frac{\pi}{6})^t$, so $S_Y = S_{\pi/2}$, $S_Z = S_{\pi/3}$ and $S_Y \circ S_Z = R_{\pi/2-\pi/3} = R_{\pi/6}$. □

Exercise 9.5.13

$R_\theta \circ S_\eta = S_{\theta+\eta}$ and $S_\eta \circ R_\theta = S_{\eta-\theta}$.

Solution. Direct multiplication with the addition formulas: the first column of $R_\theta S_\eta$ is $R_\theta(\cos\eta, \sin\eta)^t = (\cos(\theta+\eta), \sin(\theta+\eta))^t$, and similarly for the second column. For the second identity, $S_\eta R_\theta e_1 = S_\eta(\cos\theta, \sin\theta)^t = (\cos(\eta-\theta), \sin(\eta-\theta))^t$, etc. □

Exercise 9.5.14

For a reflection S of $\mathbb{R}^2$, compute $(S \circ R_{\pi/6} \circ S)^{2000}$.

Solution. $S \circ R_{\pi/6} \circ S = R_{-\pi/6}$ (Observation 9.5.9, $S = S^{-1}$). So the power is $R_{-2000\pi/6} = R_{-1000\pi/3}$, and $-\frac{1000\pi}{3} \equiv \frac{2\pi}{3} \pmod{2\pi}$ ($1000 = 3 \cdot 334 - 2$). The answer is $R_{2\pi/3}$. □

Exercise 9.5.15

Every element of $\mathbf{SO}(2)$ is a product of two reflections, but not uniquely.

Solution. By Example 9.5.11, $S_{\theta+\alpha} \circ S_\alpha = R_\theta$ for every $\alpha \in \mathbb{R}$; different $\alpha \pmod{2\pi}$ give different factorizations. □

The great principle of the plane: flip, rotate, flip back = rotate backwards.

9.6 $\mathbf{SO}(3)$ and $\mathbf{SO}(n)$

Motivation. A rotation of $\mathbb{R}^3$ has an axis and turns the perpendicular plane. Every element of $\mathbf{SO}(3)$ has eigenvalue 1 and is such a rotation, so composites of rotations are rotations. This suggests defining a rotation of $\mathbb{R}^n$ as an element of $\mathbf{SO}(n)$.

Definition 9.6.1 (Rotation of $\mathbb{R}^3$)

Let $Z \in \mathbb{R}^3$ be a unit vector and $\mathfrak{B} = \{Z, X, Y\}$ an orthonormal basis of $\mathbb{R}^3$ (so $\{X, Y\}$ is an orthonormal basis of $\langle Z\rangle^\perp$). Then $(Z, X, Y) \in \mathbf{O}(3)$ has determinant ± 1; order X, Y so that $\det(Z, X, Y) = \det(X, Y, Z) = 1$. The **rotation** $R_{Z,\theta} : \mathbb{R}^3 \to \mathbb{R}^3$ is the linear map with

$$[R_{Z,\theta}]^{\mathfrak{B}}_{\mathfrak{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} = \operatorname{diag}(1, R_\theta);$$

Z is the **axis**, $\langle X\rangle \oplus \langle Y\rangle$ the **plane of rotation**, θ the **angle**.

Equivalently (linear extension theorem), $R_{Z,\theta}$ is the unique linear map with $R_{Z,\theta}(Z) = Z$, $R_{Z,\theta}(X) = \cos\theta X + \sin\theta Y$, $R_{Z,\theta}(Y) = -\sin\theta X + \cos\theta Y$. Its independence of the choice of $\{X, Y\}$ is Exercise 9.6.6. As in §9.4,

$$[R_{Z,\theta}]^{\mathcal{E}}_{\mathcal{E}} = [I]^{\mathfrak{B}}_{\mathcal{E}} \cdot \operatorname{diag}(1, R_\theta) \cdot [I]^{\mathcal{E}}_{\mathfrak{B}},$$

where $\operatorname{diag}(1, R_\theta) \in \mathbf{SO}(3)$ and $[I]^{\mathcal{E}}_{\mathfrak{B}} = (Z, X, Y) \in \mathbf{SO}(3)$.

Observation 9.6.2

Rotations $R_{Z,\theta}$ of $\mathbb{R}^3$ belong to $\mathbf{SO}(\mathbb{R}^3)$.

Proof. By the displayed product all three factors lie in $\mathbf{SO}(3)$ (Exercise 9.3.10). ■

Question 9.6.3

Is the composite of two rotations of $\mathbb{R}^3$ a rotation? (The question is not obvious when the axes differ.)

Theorem 9.6.4

Every element of $\mathbf{SO}(\mathbb{R}^3)$ is a rotation of $\mathbb{R}^3$.

Proof. (i) *An axis.* Let $L_A \in \mathbf{SO}(\mathbb{R}^3)$. We need a unit vector Z with $AZ = Z$, i.e. $\det(A - I) = 0$. Since $\det A = \det A^t = 1$ and $A^tA = I$,

$$\begin{aligned}\det(A - I) &= \det(A^t)\det(A - I) = \det(A^t(A - I)) = \det(I - A^t)\\ &= \det((I - A)^t) = \det(I - A) = (-1)^3\det(A - I),\end{aligned}$$

so $\det(A - I) = 0$.

(ii) *The plane.* Choose an orthonormal basis $\mathfrak{B} = \{Z, X, Y\}$ with $\det(Z, X, Y) = 1$. Then $[L_A]^{\mathfrak{B}}_{\mathfrak{B}} = [I]^{\mathcal{E}}_{\mathfrak{B}}\, A\, [I]^{\mathfrak{B}}_{\mathcal{E}} \in \mathbf{SO}(3)$ and has first column e_1: $[L_A]^{\mathfrak{B}}_{\mathfrak{B}} = \left(\begin{smallmatrix}1 & a & b\\ 0 & * & *\\ 0 & * & *\end{smallmatrix}\right)$, with lower right 2×2 block B. For its columns to be orthonormal, $a = b = 0$ and $B \in \mathbf{O}(2)$; $\det B = 1$, so $B \in \mathbf{SO}(2)$ and $B = R_\theta$ (Theorem 9.5.2). ■

Second proof of (i). If $A \in \mathbf{O}(3)$ has a real eigenvalue λ with eigenvector X, then $\langle X, X\rangle = \langle AX, AX\rangle = \lambda^2\langle X, X\rangle$, so $\lambda = \pm 1$. Triangularizing A over $\mathbb{C}$ (§7.3), its eigenvalues satisfy $\lambda_1\lambda_2\lambda_3 = \det A = 1$ (Corollary 7.3.3). A real cubic has a real root, say $\lambda_1 = \pm 1$. If $\lambda_2, \lambda_3 \notin \mathbb{R}$, then $\lambda_3 = \bar{\lambda}_2$ and $\lambda_2\lambda_3 = |\lambda_2|^2 > 0$, so $\lambda_1 = 1$. If all are real, they are ± 1 with product 1, so one of them is 1. ■

Corollary 9.6.5

The composite of two rotations of $\mathbb{R}^3$ is a rotation.

Proof. By Observation 9.6.2 and Exercise 9.3.10 the composite lies in $\mathbf{SO}(\mathbb{R}^3)$; apply Theorem 9.6.4. ■

Finding the axis and angle (Z_3, θ_3) of $R_{Z_1,\theta_1} \circ R_{Z_2,\theta_2}$ means writing $R_{Z_i,\theta_i} = L_{A_i}$ and finding the unit eigenvector of A_1A_2 for the eigenvalue 1.

Exercise 9.6.6

Let $\mathfrak{B}' = \{Z, X', Y'\}$ be another orthonormal basis with $\det(Z, X', Y') = 1$, and $R'_{Z,\theta}$ the rotation defined from it. Show $R_{Z,\theta} = R'_{Z,\theta}$:
(a) since $\langle X\rangle \oplus \langle Y\rangle = \langle X'\rangle \oplus \langle Y'\rangle = \langle Z\rangle^\perp$, $[I]^{\mathfrak{B}}_{\mathfrak{B}'} = \operatorname{diag}(1, U)$ for some $U \in \mathfrak{M}_{2,2}(\mathbb{R})$;
(b) $[I]^{\mathfrak{B}}_{\mathfrak{B}'} = [I]^{\mathfrak{B}}_{\mathcal{E}}[I]^{\mathcal{E}}_{\mathfrak{B}'} \in \mathbf{SO}(3)$, hence $U \in \mathbf{SO}(2)$;
(c) $[R'_{Z,\theta}]^{\mathfrak{B}}_{\mathfrak{B}} = [I]^{\mathfrak{B}}_{\mathfrak{B}'}\operatorname{diag}(1, R_\theta)[I]^{\mathfrak{B}'}_{\mathfrak{B}} = \operatorname{diag}(1, R_\theta) = [R_{Z,\theta}]^{\mathfrak{B}}_{\mathfrak{B}}$.

Solution.

(a) Z is fixed and $X', Y' \in \langle X, Y\rangle$, so the change of coordinates has this block form.

(b) Both factors are in $\mathbf{SO}(3)$ (columns orthonormal, determinant 1), so $\operatorname{diag}(1, U) \in \mathbf{SO}(3)$, whence $U \in \mathbf{O}(2)$ with $\det U = 1$.

(c) $\operatorname{diag}(1, U)\operatorname{diag}(1, R_\theta)\operatorname{diag}(1, U^{-1}) = \operatorname{diag}(1, UR_\theta U^{-1})$, and $UR_\theta U^{-1} = R_\theta$ because $\mathbf{SO}(2)$ is commutative ($R_\alpha R_\theta = R_{\alpha+\theta}$). □

Example 9.6.7

Unlike in the plane (Theorem 9.5.7), an orthogonal operator of $\mathbb{R}^3$ need not be a rotation or a reflection: $\operatorname{diag}(-1, R_\theta) \in \mathbf{O}(3)$ has determinant -1 (not a rotation), and for $R_\theta \neq I$ it does not have 1 as an eigenvalue with a 2-dimensional eigenspace, so it is not a reflection.

Exercise 9.6.8

For $L \in \mathbf{O}(\mathbb{R}^3) - \mathbf{SO}(\mathbb{R}^3)$:
(a) L has eigenvalue -1;
(b) $[L]_{\mathfrak{B}}^{\mathfrak{B}} = \operatorname{diag}(-1, R_\theta)$ for some orthonormal basis $\mathfrak{B}$ and $\theta \in \mathbb{R}$.

Solution.

(a) With $L = L_A$, $\det A = -1$: $\det(A + I) = -\det(A^t)\det(A + I) = -\det(I + A^t) = -\det(A + I)$, so $\det(A + I) = 0$.

(b) Take a unit eigenvector Z for -1 and an orthonormal basis $\{Z, X, Y\}$. As in Theorem 9.6.4(ii), $[L]_{\mathfrak{B}}^{\mathfrak{B}} = \operatorname{diag}(-1, B)$ with $B \in \mathbf{O}(2)$ and $\det B = 1$, so $B = R_\theta$. □

So every $L \in \mathbf{O}(\mathbb{R}^3)$ is of one of three types in a suitable orthonormal frame: (1) $\operatorname{diag}(1, R_\theta)$ (a rotation); (2) $\operatorname{diag}(-1, 1, 1)$ (a reflection); (3) $\operatorname{diag}(-1, R_\eta)$ with $R_\eta \neq I$.

Unlike the plane (Theorem 9.5.2), we cannot solve the quadratic equations $A^t A = I$ directly in higher dimensions; the rest of the book finds ways around them.

Definition 9.6.9 (Rotations of $\mathbb{R}^n$)

For every n, an element of $\mathbf{SO}(\mathbb{R}^n)$ (or of $\mathbf{SO}(n)$) is called a **rotation** of $\mathbb{R}^n$.

Example 9.6.10

The 4-dimensional rotation $\operatorname{diag}(R_\theta, R_\eta)$ has two "planes of rotation" with two angles.

Describing all of $\mathbf{O}(\mathbb{R}^n)$ and $\mathbf{SO}(\mathbb{R}^n)$ is a goal of the second half of the book (Chapter 15).

Exercise 9.6.11

(a) $L \in \mathbf{O}(\mathbb{R}^n)$ iff $[L]_{\mathfrak{B}}^{\mathfrak{B}} \in \mathbf{O}(n)$ for some orthonormal basis $\mathfrak{B}$ of $\mathbb{R}^n$.
(b) The same holds with $\mathbf{SO}$ in place of $\mathbf{O}$.

Solution.

(a) Let $P = [I]_{\mathfrak{B}}^{\mathcal{E}}$, whose columns form the orthonormal basis $\mathfrak{B}$, so $P \in \mathbf{O}(n)$. Then $[L]_{\mathfrak{B}}^{\mathfrak{B}} = P^{-1}[L]_{\mathcal{E}}^{\mathcal{E}}P$, and $\mathbf{O}(n)$ is closed under products and inverses (Exercise 9.3.3). So $[L]_{\mathfrak{B}}^{\mathfrak{B}} \in \mathbf{O}(n)$ iff $[L]_{\mathcal{E}}^{\mathcal{E}} \in \mathbf{O}(n)$ iff $L \in \mathbf{O}(\mathbb{R}^n)$.

(b) Additionally $\det[L]_{\mathfrak{B}}^{\mathfrak{B}} = \det L$. □

The intuition behind this chapter is Euclidean geometry; its algebra (groups, orthonormal frames, change of frame inside $\mathbf{O}(n)$) is what we now develop in general.